

cocotb triggers

Every await in a cocotb test waits on a trigger, an object that tells the simulator when to resume your coroutine

Edge, timer, and compose triggers

- RisingEdge on a clock resumes on the next zero-to-one transition
- Timer ignores pins and completes after N time units
- First is a race: whichever trigger finishes first wins and the others are cancelled
- Combine is a join
- Pick the trigger that matches the event you actually care about

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / cocotb triggers

INTERACTIVE TOOL

cocotb triggers

Sketch **RisingEdge**, **Timer**, and **First** / **Combine** awaits on a tiny timeline. Starter:
await RisingEdge(clk) fires at t=10.

Starter example: await RisingEdge(clk) — first rise at t=10 completes the await.

Load starter example

Challenges 0 / 22 cleared

Quiz: RisingEdge: await RisingEdge(clk) resumes when...

☐ the clock next transitions 0→1

☐ any amount of time passes

☐ FallingEdge fires first always

☐ the DUT is synthesized

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

RisingEdge

Wait for next 0→1 on a pin.

Timer

Wait a duration — no edge required.

First

Earliest of several triggers wins.

Combine

Resume only when all triggers done.

Browser lab starter

learn_cocotb — cocotb triggers

Real cocotb track practice

- In the real cocotb track, open this module's examples prompts
- Restate what a trigger does in one sentence
- Sketch one await chain on paper
- Optional stretch
- Again, no live simulator required here

Pitfalls to watch

- Do not await RisingEdge when you meant FallingEdge
- Do not treat Timer like a wall-clock sleep
- First cancels losing triggers; Combine waits for all, swapping them changes test semantics
- If no edge ever arrives
- The browser timeline is simplified

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run starter RisingEdge at ten, load a First preset where the timer wins
- On paper, draw a tiny timeline with one rise, one fall
- When you are ready, take the short quiz, then continue to the cocotb clock helper